



DIU Smart Assistant: An AI-Powered University Chatbot using Retrieval-Augmented Generation (RAG)

Assignment Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	5
Level of Content	1					
Development	2					
Spelling & Grammar	1					
Organization and Formatting	1					
Total obtained mark						
Comments						

Semester: Summer 2026

Project Title:

DIU Smart Assistant: An AI-Powered University Chatbot using Retrieval-Augmented Generation (RAG)

Team Name: Bayesian Brains

Members' Names:

Student Name: Md. Gaddafi

Student ID: 232-35-678

Student Name: Md. Rafsan

Student ID: 232-35-676

Contribution:

Md. Gaddafi (232-35-678)	Literature review, requirement analysis, data collection, document preprocessing, RAG pipeline development, report writing, LLM integration, presentation preparation.
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Md. Rafsan (232-35-676)	Chatbot interface development (Streamlit), vector database integration, testing & evaluation, UI design, documentation support and deployment.
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Both we will jointly participate in project planning, implementation, debugging and final documentation.

Batch: 41

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Course Teacher Name: Md. Rashedul Alam

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1. Introduction

Artificial Intelligence (AI) has significantly transformed the way people search for information and interact with digital systems. Modern conversational AI enables users to receive accurate answers instantly through natural language communication. However, general-purpose AI chatbots may provide outdated or incorrect information when answering questions related to a specific university because they are not connected to official institutional documents.

This project proposes the development of DIU Smart Assistant, an AI-powered university chatbot based on the Retrieval-Augmented Generation (RAG) architecture. Instead of relying only on the knowledge stored in a language model, the chatbot retrieves relevant information from official university documents such as academic regulations, admission guidelines, course catalogs, academic calendars, scholarship policies, and frequently asked questions before generating responses.

The system aims to provide students, faculty members, and prospective applicants with fast, reliable, and context-aware answers while reducing the workload of administrative offices.

2. Motivation

Students frequently need information regarding admission procedures, tuition fees, course requirements, semester schedules, examination rules, scholarships, and university policies. Most of this information is scattered across multiple documents and web pages, making it time-consuming to locate.

Traditional chatbots depend on predefined rules and keywords, limiting their ability to answer complex questions. Large Language Models (LLMs) are powerful but may generate inaccurate or "hallucinated" responses if they are not grounded in trusted sources.

The motivation behind this project is to combine AI with Retrieval-Augmented Generation (RAG) to create a chatbot that retrieves information directly from official university documents before generating responses. This ensures greater accuracy, reliability, and trustworthiness while offering users a natural conversational experience.

3. Objectives

The primary objective of this project is to design and develop an intelligent AI-powered chatbot capable of answering university-related questions using Retrieval-Augmented Generation (RAG).

Specific Objectives

- Develop an AI chatbot for university information services.
- Build a knowledge base using official university documents.
- Implement semantic search using document embeddings.
- Integrate a Large Language Model (LLM) for natural language responses.
- Provide accurate, context-aware answers based on retrieved documents.
- Develop a simple and user-friendly web interface.
- Reduce the time required to search for academic information.
- Demonstrate the practical application of modern AI and NLP technologies.

4. Problem Statement

Students often face difficulties finding accurate and updated university information because relevant documents are distributed across different websites, PDF files, and notices. Searching manually is inefficient and often results in confusion.

Existing keyword-based search systems cannot fully understand natural language questions, while general AI chatbots may generate incorrect answers due to the absence of university-specific knowledge.

Therefore, there is a need for an intelligent chatbot capable of retrieving relevant information from official university documents and generating reliable, context-based responses using modern AI techniques.

5. Proposed Solution

This project proposes an AI-powered chatbot built using the Retrieval-Augmented Generation (RAG) framework.

The chatbot will first convert university documents into searchable vector embeddings and store them in a vector database. When a user submits a question, the system retrieves the most

relevant document sections through semantic similarity search. These retrieved passages are then provided to a Large Language Model, which generates an accurate and context-aware response.

The chatbot will support queries related to:

- Admission information
- Tuition fees
- Academic calendar
- Course curriculum
- Department information
- Examination regulations
- Scholarship policies
- Graduation requirements
- University facilities
- Frequently Asked Questions (FAQs)

6. Methodology / Workflow

The proposed system will follow the Retrieval-Augmented Generation (RAG) workflow.

Step 1: Data Collection

Collect official university documents such as PDF files, academic regulations, admission guidelines, course catalogs, and FAQs.

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Step 2: Data Preprocessing

Extract text, clean unnecessary content, and divide documents into smaller chunks for efficient retrieval.

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Step 3: Embedding Generation

Convert each text chunk into vector embeddings using a pre-trained embedding model.

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Step 4: Vector Database

Store all embeddings in a vector database such as ChromaDB or FAISS.



Step 5: User Query

The user enters a question through the chatbot interface.



Step 6: Semantic Retrieval

The system searches the vector database to retrieve the most relevant document chunks.



Step 7: Response Generation

The retrieved context and user query are sent to the Large Language Model (LLM), which generates a meaningful response.



Step 8: Display Response

The chatbot displays the final answer to the user.

7. Implementation Process

The implementation of the proposed system will be carried out in the following phases:

Phase 1: Requirement Analysis

- Identify user requirements.
- Define project scope and objectives.

Phase 2: Data Collection

- Gather official university documents.
- Organize knowledge sources.

Phase 3: Data Preparation

- Extract text from PDF files.
- Clean and preprocess data.
- Split documents into chunks.

Phase 4: Knowledge Base Development

- Generate embeddings.
- Store vectors in ChromaDB or FAISS.

Phase 5: AI Integration

- Connect the vector database with the Large Language Model.
- Implement the Retrieval-Augmented Generation pipeline.

Phase 6: User Interface Development

- Develop an interactive web application using Streamlit.
- Create a clean interface for user interaction.

Phase 7: Testing and Evaluation

- Test chatbot accuracy.
- Measure response relevance.
- Improve retrieval quality.

Phase 8: Documentation and Presentation

- Prepare the final report.
- Demonstrate the working system.
- Present project outcomes.

8. Expected Result

After successful implementation, the proposed chatbot is expected to:

- Provide fast and accurate responses to university-related questions.
- Improve accessibility to academic information.
- Reduce dependency on manual document searching.
- Enhance the student support experience.

- Demonstrate the practical use of Artificial Intelligence, Natural Language Processing (NLP), and Retrieval-Augmented Generation (RAG).
- Serve as a scalable prototype that can be expanded for use by educational institutions.

9. Conclusion

The proposed project, DIU Smart Assistant: An AI-Powered University Chatbot using Retrieval-Augmented Generation (RAG), aims to provide an intelligent, reliable, and user-friendly solution for accessing university information. By integrating modern AI technologies such as Natural Language Processing, vector databases, semantic search, and Large Language Models, the chatbot will deliver accurate responses based on official university documents rather than relying solely on pre-trained knowledge.

This project not only addresses a real-world problem faced by students and university staff but also provides valuable practical experience in AI application development. Furthermore, the system can be extended in the future by supporting multilingual communication, voice interaction, mobile applications, and integration with university management systems, making it a scalable solution for modern educational institutions.